Aircraft Aerodynamics Wind Tunnel Lab Report

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Abstract

This report investigates the aerodynamic characteristics of a NACA0012 aerofoil and how they vary with angle of attack. Experimental data was collected using pressure taps and a rake of Pitot tubes and processed in Excel to determine lift, drag, pitching moment, glide ratio and pressure coefficients. Conclusions based on C_l and C_d values obtained are discussed and suggestions for improvement are provided.

1 Introduction

In this lab, the aerodynamics of a NACA0012 aerofoil is investigated through wind tunnel experiments by calculating lift, drag, pitching moment, glide ratio and tunnel blockage. The main objectives are to understand the relationships between the calculated results and different angles of attack, determine the aerodynamic characteristics of the aerofoil used and to evaluate parameters that influence the accuracy of measurements. These objectives will be met by processing the mentioned data in an Excel spreadsheet, followed by producing graphs and tables for clear relationship demonstration between aerodynamic performance metrics and direct comparisons.

2 Methodology

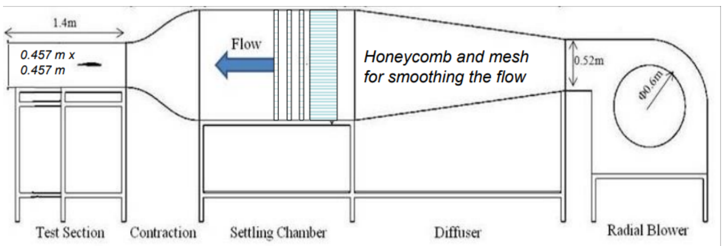


Figure 1: Schematic of the Plint Blower Wind Tunnel

These experiments were carried out in a Plint Blower Wind Tunnel shown in Figure 1, it had a test section of 0.457m x 0.457m. A NACA0012 aerofoil with a chord length of 0.152m was then placed in the test section. The aerofoil had 23 pressure taps at its mid-span location: 12 on the upper surface and 11 on the lower surface. A rake of 13 Pitot pressure probes is mounted on a traverse mechanism so that the rake can move up and down to be positioned in the wake of the flow. All the pressures are measured using pressure transducers, this data is then exported to an Excel spreadsheet. The procedure followed during this experiment is as follows:

- Initially the aerofoil's angle of attack is set to approximately zero.
- The wind tunnel can then be turned on and set to approximately 20m/s.
- Now set the actual aerofoil to incidence angle of zero by balancing the upper and lower surface pressures as seen on the display. Upper surface is represented by the green bars and the lower surface is represented by the white bars. Now press the 'Get Zero' button on the display.
- Now traverse the rake so that the lowest pressure aligns with the 0mm line on the display.
- Save the data 4 times by pressing the 'Save Data' button on the display 4 times.
- Repeat the 2 previous steps for angles of attack 3°, 6°, 9°, 12° and 15°.

Now that all the data has been collected, the first step is to average the angle of attacks and the freestream static pressures. Then average the measurements for the pressure at each tap for upper and lower surfaces. After this average the pressure measurements of the wake. From these averaged values the pressure coefficients at each point on the upper and lower surface can be calculated using the following equation:

$$C_p = \frac{p - p_\infty}{\frac{1}{2}\rho_\infty V_\infty^2} = \frac{p - p_\infty}{p_t - p_\infty} \tag{1}$$

C_p is the pressure coefficient at the point being evaluated. p is the static pressure at the point being evaluated. p_∞ is the freestream static pressure. ρ_∞ is the fluid density. V_∞ is the velocity of the freestream fluid. p_t is the total freestream pressure. The value of p_t is taken to be equal to p_{24}, which is the the rake tube furthest from the aerofoil. After this the integration of the upper C_{p_u} and lower C_{p_l} surfaces should be done to attain the coefficient of the normal force C_n acting on the aerofoil. The equation to do this is shown below, however within the Excel it has carried out using the trapezium rule.

$$C_n = \frac{1}{c}\oint_{Lower} C_p dx - \frac{1}{c}\oint_{Upper} C_p dx \tag{2}$$

c is the chord length. Within the Excel it was calculated by finding the difference between the two pressure tap locations next to each other. This was when performing the trapezium rule.

As the angle of attack is relatively small, it can be assumed that the coefficient of lift is only dependant on the force normal to the aerofoil surface. Therefore, the equation for the coefficient of lift is given as:

$$C_l = C_n cos\alpha \tag{3}$$

To calculate the drag of the area must construct a control volume around the aerofoil. It is assumed that the flow is steady and p_∞ is acting at boundaries of control volume. This leads to the equation below, which must be computed using the trapezium in Excel:

$$D' = \int_d^b \rho u_2(u_1 - u_2)dy = \int_{wake} \rho u(y)(V_\infty - u(y))dy \tag{4}$$

ρ is the density of the fluid. u_1 and u_2 is the fluid velocity in and out of the control volume. The points b and d are the streamlines at the exit of the control volume. $u(y)$ can be calculated using readings from the rake and solving for dynamic pressure. Then dividing by dynamic pressure coefficient of drag c_d can be calculated using the following equation:

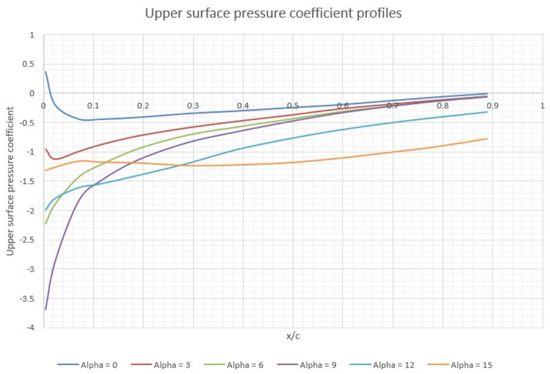
$$c_d = \frac{D'}{\frac{1}{2}\rho V_\infty^2 c} \tag{5}$$

Finally, to compute the coefficient of moment about the quarter chord the following integral would be evaluated using the trapezium rule within the Excel:

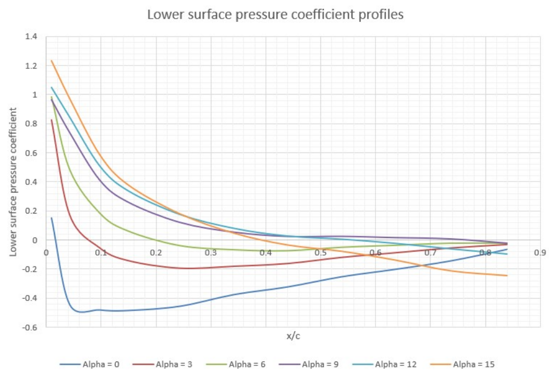
$$C_m = \frac{1}{c^2}\oint_{Lower} C_p(x - x_{ref})dx - \frac{1}{c^2}\oint_{Upper} C_p(x - x_{ref})dx \tag{6}$$

The data in the Excel is given as $\frac{x}{c}$. When the coefficient of moment is positive this means that the aircraft will pitch up. [2]

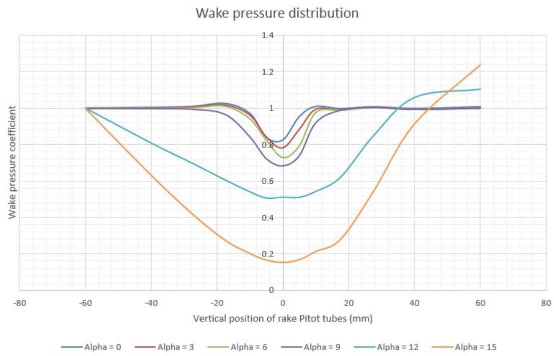
3 Results & Discussion



(a) Upper surface c_{ps} profiles



(b) Lower surface c_{ps} profiles



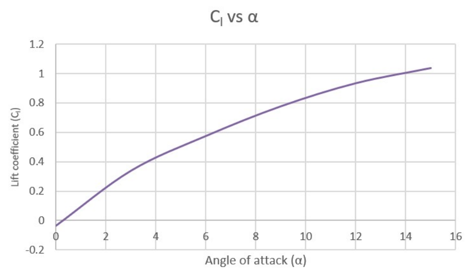
(c) c_{pw} distributions

Figure 2: Pressure coefficients for different angles of attack

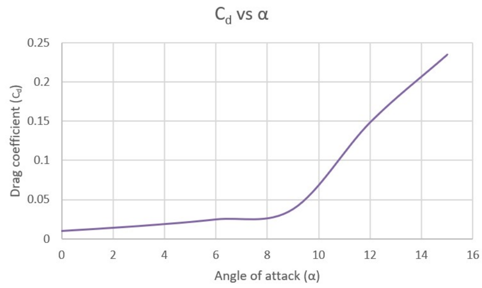
In Figure 2a, it can be seen that only for an angle of attack of 0, c_{ps} is positive at the leading edge. This is because for a symmetric aerofoil at 0°, there is a very weak suction peak at the leading edge due to the location of the stagnation point being at $\frac{x}{c} = 0$. There is a suction peak for the other angles of attack which is why those lines begin at negative c_{ps} values. However for larger angles of attack (12° and 15°), there is a very steep adverse pressure gradient which brings the point of flow separation closer to the leading edge, decreasing the effect of the suction peak and making the leading edge c_{ps} value less negative. As can be seen, all of the lines gradually tend close to 0 as the flow moves towards the trailing edge. This demonstrates pressure recovery and this is less effective for higher angles of attack due to flow separation occurring earlier.

In Figure 2b, the general trend is that c_{ps} values are positive at the leading edge and tend to be close to 0 towards the trailing edge. This is because the lower surface experiences compression due to the flow deflecting against the aerofoil, with the strength of this compression increasing with angle of attack, making the c_{ps} values more positive at higher angles of attack at the leading edge. As flow accelerates towards the trailing edge, the pressure gradually decreases. Since the aerofoil used was symmetrical, at 0°, the flow is divided evenly between the upper and lower surfaces, causing the deflection at the leading edge to be much less positive than higher angles of attack.

In Figure 2c, it can be seen that for angles of attack of 0°, 3°, 6° and 9°, the wake is narrow and somewhat symmetrical, as higher drag is associated with higher angles of attack. Due to flow separation, the wake pressure distribution becomes less symmetric for higher angles of attack. A value of 0 for vertical position of the rake Pitot tubes corresponds to the centreline of the wake. There is a dip at this point due to the high velocity deficit which causes a decreases in static pressure aft of the trailing edge, which in turn decreases c_{pw}. At angles of attack of 12° and 15°, the pressure cannot recover back to the freestream pressure like the other angles of attack due to a turbulent wake causing high energy loss. Overall, as angles of attack increases, c_{pw} decreases.



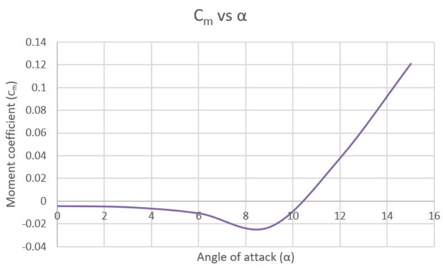
(a) Lift coefficient vs α



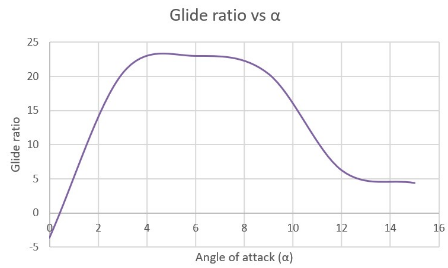
(b) Drag coefficient vs α

Figure 3

In Figure 3a, as α increases, C_l increases. As angle of attack increases, velocity of flow over the upper surface increases which leads to an increase in momentum. This creates a higher pressure difference between the upper and lower surfaces, generating lift. For a symmetric aerofoil at 0°, C_l should be 0, however the graph shows a small negative value at this point which could be due to the aerofoil not being perfectly symmetrical or small errors with the equipment. In Figure 3b, the value of C_d only starts to increase significantly after an angle of attack of about 9°. Before this point, the flow remains fully attached to the aerofoil but between 9° and 15°, the flow in the boundary layer transitions from laminar to turbulent, leading to higher skin friction drag and therefore contributing to a steeper gradient on the graph. Figure 3 shows no evidence that stall has already occurred, however, it is likely approaching.



(a) Moment coefficient vs α



(b) Glide ratio vs α

Figure 4

In Figure 4a, C_m remains roughly constant near 0 until an angle of attack of about 6°. This is because for a symmetric aerofoil the pressure distribution is also symmetric for low angles of attack, therefore a pitching moment is not generated about the quarter chord point. At 6° to 10 °, this pressure distribution changes, causing the centre of pressure to move forward. This generates a negative pitching moment, which is shown by the slight dip on the graph in this range of angles of attack. After 10°, C_m increases with angle of attack at a steady rate. A greater pressure recovery and therefore stronger adverse pressure gradient can be observed at higher angles of attack. This results in more lift acting further along the chord line of the aerofoil, moving the centre of pressure back. This generates a positive pitching moment.

In Figure 4b, glide ratio increases steadily between 0° and 4° because as angle of attack increases, both lift and drag increase, with lift increasing at a faster rate than drag. Glide ratio stays roughly constant between 4° and 8° because here, lift and drag are both increasing. For higher angles of attack, it can be seen that glide ratio decreases due to flow separation causing drag to increase faster than lift. The graph levels off between 12° and 15° because the aerofoil is reaching the stall point and the lift to drag ratio is roughly constant which leads to a roughly constant glide ratio.

Angle of attack	**Experimental C_l**	**Analytical C_l**	**Thin aerofoil C_l**	**Experimental C_d**	**Analytical C_d**
0	-0.03690373	0	0	0.010238231	0.013
3	0.337360724	0.449	0.328986813	0.016521517	0.0121
6	0.57384257	0.691	0.657973627	0.024979108	0.0168
9	0.776072447	0.924	0.98696044	0.038131718	0.0287
12	0.933227303	1.016	1.315947253	0.148856463	0.0558
15	1.036508016	0.685	1.644934067	0.235286703	0.1749

Table 1

Table 1 highlights that the analytical results for C_l are very similar to the thin aerofoil theory results for C_l (where $C_l = 2\pi\alpha$) but are different to the experimental C_l values obtained. Analytical methods and thin aerofoil theory assume inviscid flow, therefore viscous effects in the boundary layer are not considered. There could also be small imperfections with the setup of the wind tunnel and the aerofoil used could have had a slightly rough surface that affects the aerodynamics and in turn the experimental calculation of C_l. The experimental and analytical results for C_d are similar for low angles of attack, however there is a significant difference for higher angles of attack. The slower increase in analytical C_d values could be due to the analysis neglecting the effects of flow separation and not being able to model complex flows.

The motivation for tunnel corrections is to be able to better simulate results obtained in free air by minimising the effects of solid and wake tunnel blockage [1]. Using equations given in the lab handbook [2], values for total blockage ratio were obtained (with solid blockage, $\epsilon_{sb} = 0.04646266$):

Angle of attack	**Wake Blockage (ϵ_{wb})**	**Total Blockage Effect (ϵ)**
0	0.001702638	4.816529755
3	0.002747561	4.921022047
6	0.004154075	5.061673447
9	0.00634138	5.280403945
12	0.024755123	7.121778251
15	0.039128642	8.559130178

Table 2

Table 2 shows that tunnel corrections are required for all angles of attack other than 0° and 3° because ϵ is less than 5% for only these two configurations.

4 Conclusions

This report analysed the aerodynamic properties of a NACA0012 aerofoil and how they varied with angle of attack. To measure these aerodynamic properties, pressure taps and a rake of Pitot tubes were used. These pressure readings then allowed for the calculation of pressure, lift and drag coefficients and glide ratio for different angles of attack. These calculations were carried out in Excel using approximation methods like the trapezium rule.

A key finding was that lift coefficient increased with angle of attack at a decreasing rate. This will ultimately lead to the aerofoil displaying stall properties at large angles of attack. Drag coefficient drastically increased at about 9°, this is the pressure drag increasing. This is because at this point, flow separation occurs, resulting in nearing stall of the aircraft. Up until 9° the drag force will predominantly come from skin friction drag.

CFD and experimental results tend to align for small angles of attack and diverge for greater angles of attack. This is often due to the limitations of a theoretical model. Pitching moment at higher angles of attack becomes increasingly negative which results in the pitching down of the nose, which can be extremely dangerous for aircraft. Curvature effects would need to be taken into account, as tunnel adjustments would be negligible. The experimental setup could be improved to account for temperature and pressure changes. Additionally, if more results were taken for more angles of attack this would indicate a clearer picture of the stall characteristics of the aerofoil. Increasing the resolution of the pressure taps and rake would provide more accurate results.

References

[1] AERO31321 Aircraft Aerodynamics Wind Tunnel Lab Presentation
[2] AERO31321 Aircraft Aerodynamics Wind Tunnel Lab Handbook